

Computational Analysis of UCSF Clinical Trial Data Using MySQL, Natural Language Processing, and Machine Learning

William Lorenzo^{1*}

^{1*}Department of Statistics, B.S.: Data Science Concentration,
California State University, East Bay, 25800 Carlos Bee Boulevard,
Hayward, California, 94542, USA.

Corresponding author(s). E-mail(s): wlorenzo@horizon.csueastbay.edu;

Abstract

Clinical trials generate structured and semi-structured data that can support biomedical informatics, clinical operations research, and healthcare analytics. This study presents a computational workflow for analyzing UCSF-affiliated clinical trial records obtained from ClinicalTrials.gov. The project integrates a MySQL relational database with Python-based extract-transform-load processing, exploratory data analysis, natural language processing, and machine learning. Clinical trial metadata were normalized into trial, condition, and intervention tables using one-to-many relational structures. SQL queries were used to summarize trial phases, sponsors, enrollment patterns, medical conditions, and intervention categories. Python was then used to visualize enrollment distributions, identify highly represented conditions, analyze intervention types, and evaluate phase-level enrollment differences. Text-based machine learning methods were also applied to study titles using term frequency-inverse document frequency features. Logistic regression was used as a supervised baseline for predicting trial phase, while K-means clustering was applied to group trials by title similarity. Results showed that Phase 3 trials had substantially larger enrollment than earlier-phase trials, drug interventions dominated the intervention landscape, and UCSF served as the central sponsor for the majority of studies. The machine learning analysis demonstrated that short clinical trial titles contain meaningful textual patterns but are limited for phase prediction when class distributions are imbalanced. Overall, the study demonstrates how relational databases, SQL analytics, Python visualization, and machine learning can be combined to support clinical trial informatics using real-world public healthcare data.

Keywords: clinical trial informatics, ClinicalTrials.gov, MySQL, Python, healthcare analytics, TF-IDF, logistic regression, K-means clustering, biomedical data science, UCSF

1 Introduction

Clinical trials are essential for evaluating the safety, efficacy, and clinical value of medical interventions. They provide evidence for drug development, device testing, behavioral interventions, diagnostic procedures, and treatment protocols. Because clinical trial registries contain structured fields such as study phase, enrollment, sponsor, intervention type, condition, location, and outcome measures, they are valuable sources for computational healthcare analytics.

ClinicalTrials.gov is a public registry and results database maintained by the U.S. National Library of Medicine. It contains clinical study records submitted by academic institutions, hospitals, government agencies, biotechnology companies, and pharmaceutical sponsors. These records are especially useful for data science projects because they contain real-world medical research metadata rather than synthetic or simplified examples.

The objective of this project was to build a reproducible computational workflow for analyzing UCSF-affiliated clinical trial data using MySQL and Python. The workflow combined relational database design, Python-based preprocessing, SQL analytics, visualization, and machine learning. The analysis focused on completed interventional trials with available results, allowing the project to examine trial phases, enrollment sizes, sponsors, conditions, and intervention categories.

This study also demonstrates the role of natural language processing in biomedical informatics. Clinical trial titles were transformed into numerical features using term frequency-inverse document frequency. These features were used in both supervised and unsupervised machine learning workflows. Logistic regression was used to evaluate whether study titles could predict trial phase, while K-means clustering was used to group trials based on textual similarity.

2 Dataset Source and Study Design

The dataset was downloaded from ClinicalTrials.gov using filters designed to retrieve UCSF-affiliated clinical trials. The search focused on completed interventional studies with results and included records associated with the University of California, San Francisco either as sponsor, collaborator, or clinical trial location. The exported file was named `ctg-studies.csv`.

The dataset contained 103 clinical trial records and 23 columns. Key variables included the National Clinical Trial number, study title, study URL, study status, conditions, interventions, primary and secondary outcome measures, sponsor, collaborators, sex, age group, phase, enrollment, funder type, study type, study design, dates, and locations.

The National Clinical Trial identifier, denoted as `nct_number`, was used as the primary key for each study. Several fields, including conditions and interventions, contained multiple values separated by delimiters. These were normalized into separate relational tables in MySQL.

3 MySQL Database Architecture

The MySQL database was designed to represent clinical trial metadata using relational structures. The primary table was the `trials` table, where each row represented one clinical trial. The `conditions` and `interventions` tables were created to represent one-to-many relationships because one trial can study multiple medical conditions and can include multiple intervention types.

Let T denote the set of clinical trials, C denote the set of medical conditions, and I denote the set of interventions. The relational design can be represented as

$$T \rightarrow C, \quad T \rightarrow I, \quad (1)$$

where each trial $t_i \in T$ may be associated with multiple conditions $c_{ij} \in C$ and multiple interventions $i_{ik} \in I$.

The one-to-many relationship between trials and conditions is defined by the foreign key

$$\text{conditions.nct_number} \rightarrow \text{trials.nct_number}. \quad (2)$$

Similarly, the relationship between trials and interventions is defined by

$$\text{interventions.nct_number} \rightarrow \text{trials.nct_number}. \quad (3)$$

This structure reduces redundancy and supports efficient SQL aggregation. For example, medical conditions can be counted independently of trial-level metadata, and intervention types can be summarized across all studies.

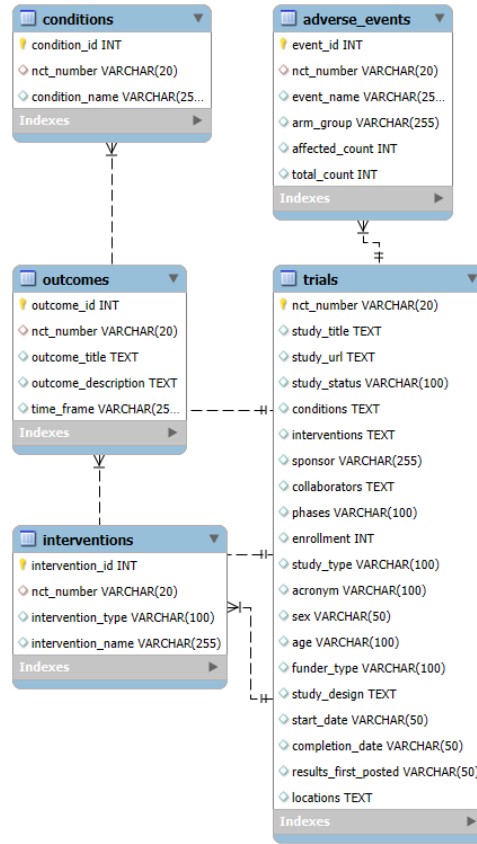


Fig. 1 Enhanced entity-relationship diagram for the UCSF clinical trial MySQL database. The schema shows the central `trials` table and its one-to-many relationships with the `conditions`, `interventions`, `outcomes`, and `adverse_events` tables through the `nct_number` foreign key.

3.1 Database schema

The database contained the following core tables:

- **trials**: stores study-level metadata.
- **conditions**: stores normalized medical conditions linked to trial IDs.
- **interventions**: stores normalized intervention types and names linked to trial IDs.
- **outcomes**: reserved for future ClinicalTrials.gov outcome extraction.
- **adverse_events**: reserved for future adverse event extraction.

4 Python ETL Pipeline

The Python extract-transform-load pipeline was developed using pandas and SQLAlchemy. First, the exported ClinicalTrials.gov CSV file was loaded into a pandas DataFrame. Column names were converted from their original ClinicalTrials.gov format into SQL-friendly names using lowercase characters and underscores. The enrollment column was converted into an integer variable, with non-numeric or missing values treated as zero.

The trial-level variables were inserted into the MySQL `trials` table. The conditions field was split using the vertical bar delimiter, and each condition was inserted as a separate row into the `conditions` table. The interventions field was also split using the same delimiter. Each intervention was parsed into an intervention type and intervention name using the colon separator.

Algorithm 1 summarizes the ETL process.

Algorithm 1 Python and MySQL ETL workflow for UCSF clinical trial data

```
1: Load ctg-studies.csv into a pandas DataFrame
2: Rename columns using SQL-compatible naming conventions
3: Convert enrollment into an integer variable
4: Select trial-level variables and insert them into the MySQL trials table
5: for each clinical trial record do
6:     Split the conditions field into individual condition values
7:     Insert each condition into the conditions table with its associated nct_number
8: end for
9: for each clinical trial record do
10:    Split the interventions field into individual intervention values
11:    Parse each intervention into type and name
12:    Insert each intervention into the interventions table with its associated
        nct_number
13: end for
14: Validate table counts using SQL queries
```

5 Statistical Analysis

Enrollment size was treated as the main quantitative variable. For a sample of n clinical trials with enrollment values $x_1, x_2, \dots, x_n$, the sample mean enrollment is defined as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (4)$$

The sample standard deviation is defined as

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (5)$$

Because clinical trial enrollment was highly right-skewed, a logarithmic transformation was applied:

$$z_i = \log(1 + x_i), \quad (6)$$

where x_i is the original enrollment value for trial i . The addition of 1 prevents undefined values when enrollment is zero.

6 Exploratory Data Analysis

Exploratory data analysis was performed using SQL queries and Python visualizations. The analysis focused on condition frequency, intervention type distribution, average enrollment by phase, sponsor representation, enrollment distribution, and log-transformed enrollment.

6.1 Summary statistics for enrollment

The dataset contained 103 trials with positive enrollment values. The mean enrollment was 128.37 participants, while the median enrollment was 43 participants. The standard deviation was 446.45 participants, indicating substantial variability across studies. The minimum enrollment was 2 participants and the maximum enrollment was 4,446 participants.

Table 1 summarizes the enrollment distribution.

Table 1 Summary statistics for clinical trial enrollment

Statistic	Enrollment
Count	103
Mean	128.37
Standard deviation	446.45
Minimum	2
First quartile	24
Median	43
Third quartile	87.5
Maximum	4446

The descriptive statistics indicate that enrollment sizes varied substantially across the UCSF-affiliated clinical trial dataset. The average enrollment was larger than the median, which suggests that a small number of large trials increased the overall mean. Most studies enrolled fewer than 100 participants, while a few large studies enrolled hundreds or thousands of participants.

6.2 Enrollment by clinical trial phase

Table 2 summarizes enrollment by phase. Phase 2 trials represented the largest group, with 68 studies. Phase 3 trials had the highest average enrollment, with a mean of 542.73 participants and a maximum enrollment of 4,446 participants.

Table 2 Enrollment statistics by clinical trial phase

Phase	Count	Mean	Median	Maximum
PHASE1 PHASE2	15	30.53	27.0	67
PHASE2	68	56.07	40.0	294
PHASE2 PHASE3	5	162.00	98.0	385
PHASE3	15	542.73	216.0	4446

The phase-level analysis demonstrates a clear relationship between trial phase and enrollment size. Phase 3 studies had the largest average enrollment because they are typically designed to confirm treatment efficacy and safety in broader populations. In contrast, Phase 1/Phase 2 trials had the smallest enrollment because early-stage studies are often exploratory and safety-focused.

6.3 Exploratory figures

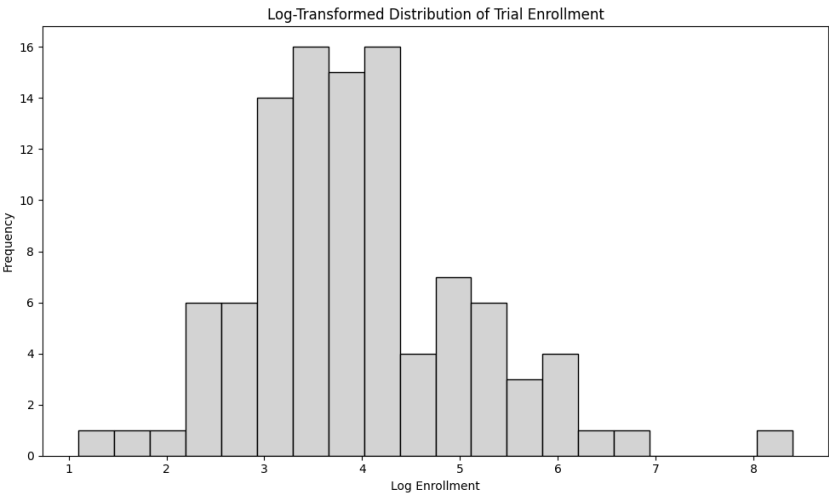


Fig. 2 Log-transformed distribution of clinical trial enrollment.

The log-transformed histogram reduces the impact of extremely large studies and makes the structure of the enrollment distribution easier to observe. Most trials cluster

around moderate log-enrollment values, while a smaller number remain at the upper end of the distribution.

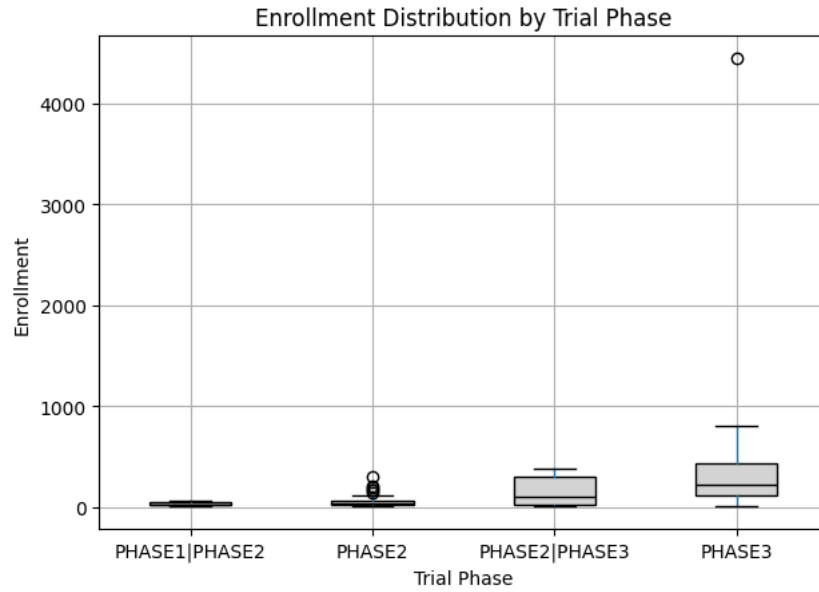


Fig. 3 Enrollment distribution by clinical trial phase.

The boxplot shows that enrollment generally increases as trials move into later phases. Phase 3 trials display the largest spread and include a major high-enrollment outlier.

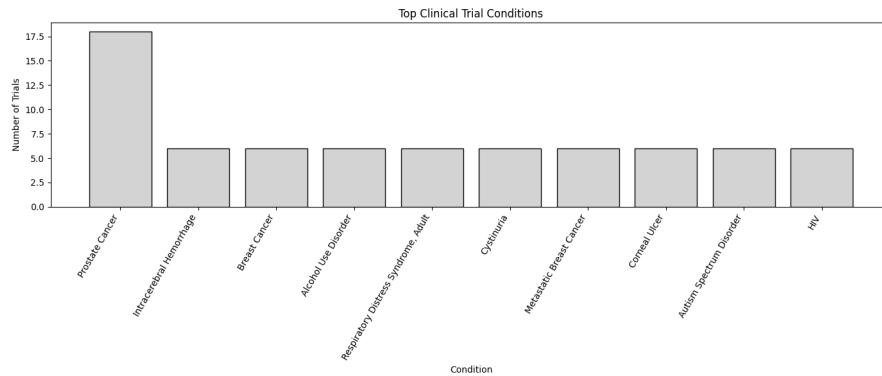


Fig. 4 Top clinical trial conditions in the UCSF-affiliated dataset.

The most common medical condition was prostate cancer. Other conditions appearing multiple times included breast cancer, alcohol use disorder, HIV, autism spectrum

disorder, intracerebral hemorrhage, and corneal ulcer. The distribution indicates that the dataset spans multiple clinical domains, including oncology, infectious disease, neurology, psychiatry, and ophthalmology.

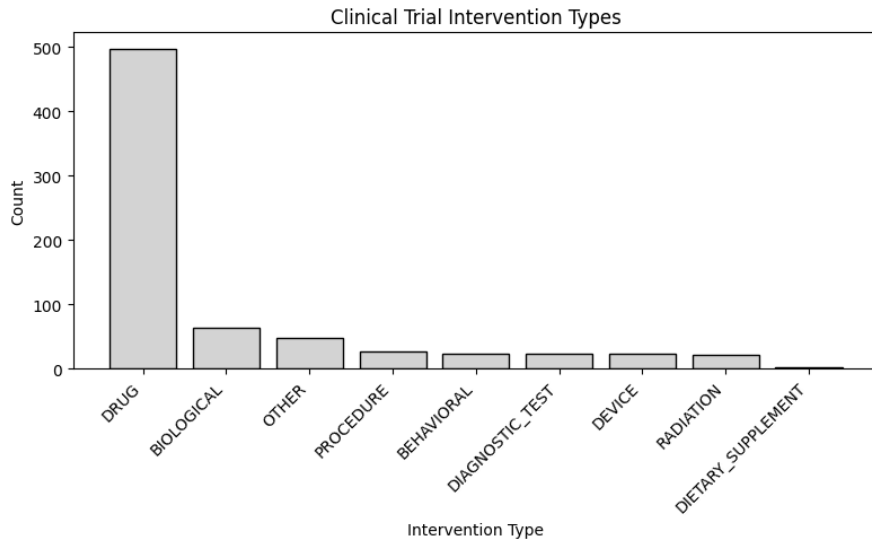


Fig. 5 Distribution of intervention types across clinical trials.

Drug interventions dominated the dataset, with substantially more occurrences than biological, procedural, behavioral, device-based, radiation-based, diagnostic, or dietary interventions. This pattern is consistent with the central role of pharmaceutical interventions in clinical research.

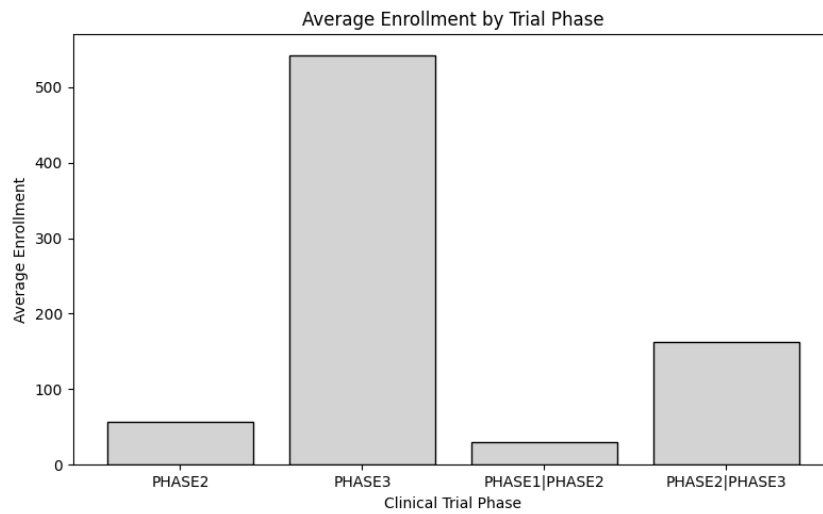


Fig. 6 Average enrollment by clinical trial phase.

Average enrollment was highest for Phase 3 trials and lowest for Phase 1/Phase 2 trials. This reflects the clinical trial development pathway, where later-phase studies generally require larger sample sizes.

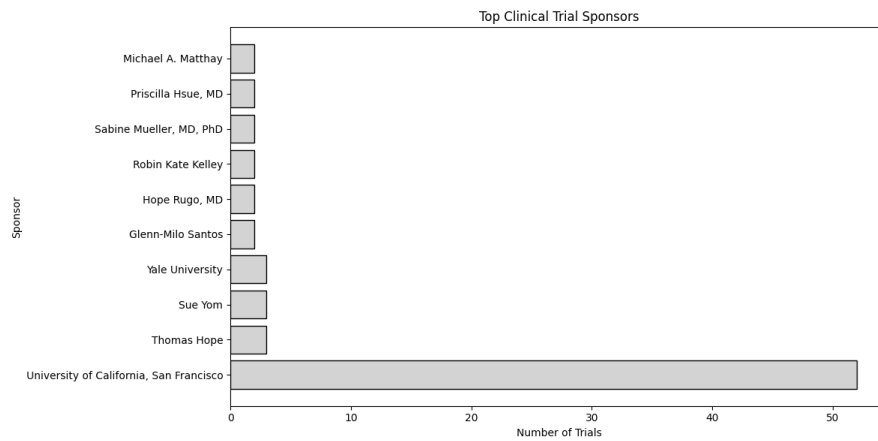


Fig. 7 Top clinical trial sponsors in the UCSF-affiliated dataset.

The University of California, San Francisco was the dominant sponsor in the dataset. The remaining sponsors contributed a much smaller number of studies, indicating that the dataset is strongly centered on UCSF-led or UCSF-affiliated clinical research.

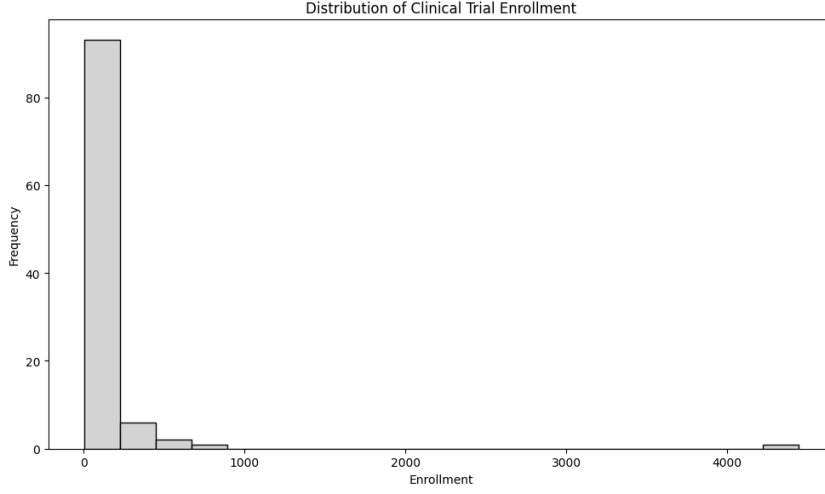


Fig. 8 Original distribution of clinical trial enrollment.

The original enrollment distribution was highly right-skewed. Most trials enrolled relatively small numbers of participants, while a few large studies produced a long right tail.

7 Natural Language Processing and Machine Learning

Clinical trial titles were analyzed using term frequency-inverse document frequency. For term t in document d , the TF-IDF value is defined as

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \log \left(\frac{N}{\text{df}(t)} \right), \quad (7)$$

where $\text{tf}(t, d)$ is the frequency of term t in document d , N is the total number of documents, and $\text{df}(t)$ is the number of documents containing term t .

7.1 Logistic regression for trial phase prediction

A logistic regression model was trained to predict clinical trial phase from study title text. For binary logistic regression, the probability of class 1 is modeled as

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}. \quad (8)$$

For multi-class classification, logistic regression generalizes to the softmax function:

$$P(y = k \mid x) = \frac{e^{\beta_k^T x}}{\sum_{j=1}^K e^{\beta_j^T x}}, \quad (9)$$

where K is the number of phase categories.

The model achieved an overall accuracy of 0.69 on the test set. However, the classification report showed that the model primarily predicted the majority class, Phase 2. The test set contained 18 Phase 2 examples out of 26 total examples, producing an imbalanced classification setting. The model achieved strong recall for Phase 2 but failed to identify smaller phase categories such as Phase 1/Phase 2, Phase 2/Phase 3, and Phase 3.

Table 3 summarizes the logistic regression classification results.

Table 3 Logistic regression classification report for trial phase prediction

Class	Precision	Recall	F1-score	Support
PHASE1 PHASE2	0.00	0.00	0.00	3
PHASE2	0.69	1.00	0.82	18
PHASE2 PHASE3	0.00	0.00	0.00	1
PHASE3	0.00	0.00	0.00	4
Accuracy			0.69	26
Macro average	0.17	0.25	0.20	26
Weighted average	0.48	0.69	0.57	26

The results indicate that study title text alone is insufficient for reliable phase prediction in this small and imbalanced dataset. Additional predictors, including enrollment, intervention type, funder type, sponsor, and study design, would likely improve predictive performance.

7.2 K-means clustering of clinical trial titles

K-means clustering was used to group clinical trials according to title similarity. Given K clusters, K-means minimizes the within-cluster sum of squares:

$$\arg \min_S \sum_{k=1}^K \sum_{x_i \in S_k} \|x_i - \mu_k\|^2, \quad (10)$$

where S_k is cluster k and μ_k is the centroid of that cluster.

In this project, $K = 4$ clusters were used. The model grouped study titles into four text-based categories. Some clusters appeared to contain oncology-related studies involving cancer therapies, radiation treatment, and drug combinations. Other clusters included studies involving neurological disorders, chronic diseases, ophthalmology, and general medical interventions.

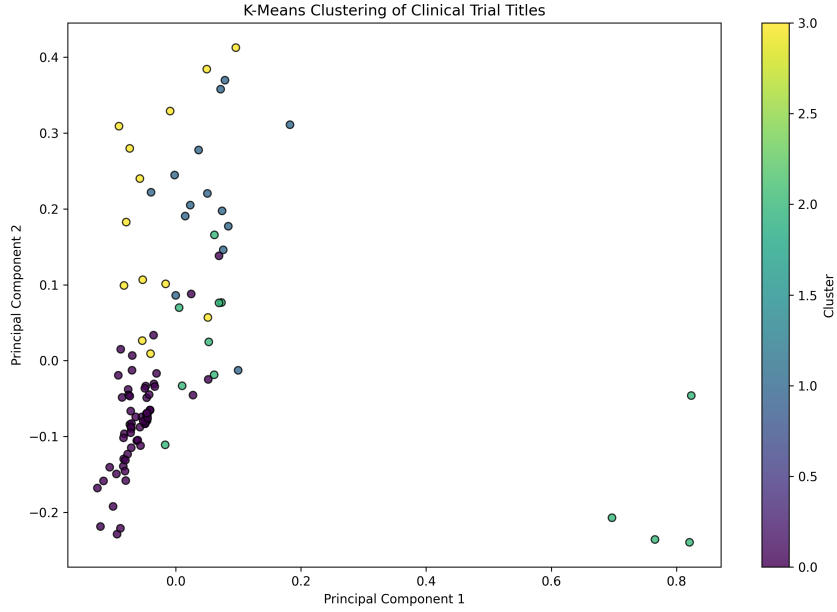


Fig. 9 K-means clustering output for clinical trial titles.

The clusters should be interpreted as exploratory rather than definitive clinical categories. Clinical trial titles are short, and many studies overlap across medical specialties. Nevertheless, the clustering analysis demonstrates that unsupervised machine learning can identify meaningful textual patterns in clinical research metadata.

8 Results

The MySQL database successfully stored and normalized clinical trial metadata into trial, condition, and intervention tables. The relational design allowed SQL aggregation across conditions, intervention types, sponsors, phases, and enrollment values.

The exploratory analysis showed that enrollment sizes were highly variable and right-skewed. The mean enrollment was 128.37 participants, while the median was 43 participants. This difference indicates the influence of a small number of very large trials. Phase 3 trials had the highest average enrollment, consistent with their role in confirmatory treatment evaluation.

Drug interventions represented the largest intervention category, indicating that pharmaceutical treatments were the dominant research focus. Biological therapies, procedures, devices, behavioral interventions, radiation therapy, and diagnostic interventions appeared less frequently. This pattern reflects the structure of many real-world clinical trial portfolios, where drug-based trials are highly represented.

The condition analysis showed that prostate cancer appeared most frequently among the top conditions, followed by several conditions that occurred multiple times.

These included breast cancer, alcohol use disorder, HIV, autism spectrum disorder, intracerebral hemorrhage, and corneal ulcer. The dataset therefore represented a diverse collection of clinical domains.

The supervised machine learning model demonstrated limited performance for phase prediction. Although the model achieved 0.69 accuracy, this was largely due to correct predictions of the majority Phase 2 class. The model did not generalize well to underrepresented phase categories. The clustering model provided exploratory groupings of clinical trial titles and identified some thematic patterns, particularly around oncology and treatment-focused trials.

9 Discussion

This project demonstrates how MySQL and Python can be integrated to support clinical trial informatics. The MySQL database provided a structured foundation for storing normalized clinical trial records, while Python provided flexible tools for pre-processing, visualization, statistical analysis, natural language processing, and machine learning.

The enrollment analysis revealed patterns consistent with clinical trial development. Early-phase and mid-phase studies tended to enroll fewer participants, while Phase 3 studies enrolled substantially larger populations. This reflects the increasing evidentiary requirements of later-stage clinical research, where larger samples are needed to evaluate effectiveness, monitor safety, and support regulatory decision-making.

The intervention analysis showed that drug-based interventions dominated the dataset. This finding is consistent with the central role of pharmaceutical research in clinical trials. However, the presence of biological, procedural, behavioral, device-based, radiation-based, and diagnostic interventions also demonstrates the diversity of clinical research activities represented in UCSF-affiliated studies.

The machine learning results highlight both the potential and limitations of text-based modeling in small biomedical datasets. TF-IDF features extracted from study titles captured meaningful terminology, but the supervised model was limited by class imbalance and the small number of observations. Logistic regression favored the majority Phase 2 class, producing misleadingly moderate accuracy while failing to predict smaller categories. This demonstrates the importance of examining precision, recall, F1-score, and class support rather than relying on accuracy alone.

K-means clustering provided a useful exploratory tool for organizing trials by title similarity. While the clusters were not perfect clinical categories, they suggested that study titles contain enough information to reveal thematic patterns. Future work could improve this analysis using larger datasets, condition text, intervention names, eligibility criteria, outcome measures, and advanced language models.

10 Limitations

Several limitations should be acknowledged. First, the dataset contained only 103 clinical trials, which limits the reliability of machine learning models. Second, the logistic regression model used study titles only, excluding potentially useful variables

such as enrollment, intervention type, sponsor, funder type, and study design. Third, ClinicalTrials.gov exports contain aggregate trial metadata rather than patient-level clinical data. Therefore, this project cannot evaluate individual treatment responses or patient-level outcomes.

Fourth, the dataset was filtered around UCSF-affiliated studies, meaning the findings may not generalize to all clinical trials. Fifth, the conditions and interventions fields required text parsing, which may introduce inconsistencies when study records use nonstandard wording. Finally, the outcomes and adverse event tables were designed but not fully populated in the present workflow. Future work should use the ClinicalTrials.gov API to extract structured outcome measures, participant flow, statistical analyses, and adverse event summaries.

11 Conclusion

This study presented a computational workflow for analyzing UCSF-affiliated clinical trial data using MySQL, Python, natural language processing, and machine learning. The project transformed a ClinicalTrials.gov CSV export into a normalized relational database, performed SQL-based analytics, created Python visualizations, and applied text-based machine learning to clinical trial titles.

The analysis showed that clinical trial enrollment was highly right-skewed, Phase 3 trials had the largest average enrollment, drug interventions were the dominant intervention type, and UCSF was the leading sponsor in the dataset. Logistic regression demonstrated that title text alone was insufficient for reliable phase prediction due to class imbalance and limited sample size. K-means clustering showed that study titles can still reveal useful thematic groupings for exploratory clinical trial analysis.

Overall, this project demonstrates the value of combining relational database systems with Python-based analytics for real-world healthcare data science. The workflow provides a foundation for future clinical trial informatics studies involving richer outcome data, adverse event reporting, eligibility criteria, and larger multi-institutional datasets.

Supplementary information. Supplementary material may include the full Python ETL script, MySQL schema, SQL analytics queries, exported visualizations, and machine learning outputs.

Acknowledgements. The author acknowledges California State University, East Bay for academic support and the instructional environment that supported this healthcare analytics and machine learning project.

Declarations

Funding

No external funding was reported for this study.

Conflict of interest/Competing interests

The author declares no competing interests.

Ethics approval and consent to participate

Not applicable. This study used publicly available clinical trial registry metadata

and did not involve direct interaction with human participants, animal subjects, or identifiable private health information.

Consent for publication

Not applicable.

Data availability

The data used in this study were obtained from ClinicalTrials.gov as a CSV export of UCSF-affiliated completed interventional studies with available results. The local dataset file was named `ctg-studies.csv`.

Materials availability

Not applicable.

Code availability

The MySQL schema, SQL queries, Python ETL code, visualization code, and machine learning code are included in Appendix C.

Author contribution

William Lorenzo contributed to dataset collection, database design, Python preprocessing, SQL analytics, visualization, machine learning analysis, interpretation of results, and manuscript drafting.

Appendix A Figures

- `top_conditions.png`
- `intervention_types.png`
- `avg_enrollment_by_phase.png`
- `top_sponsors.png`
- `enrollment_distribution.png`
- `log_enrollment_distribution.png`
- `enrollment_boxplot_by_phase.png`
- `kmeans_trial_clusters.png`

Appendix B Full MySQL Code

Listing 1 MySQL schema and analytics queries for UCSF clinical trial data

```
1  -- Computational Analysis of UCSF Clinical Trial Data Using MySQL
   , NLP, and Machine Learning
2
3  CREATE DATABASE ucsf_clinical_trials;
4  USE ucsf_clinical_trials;
5
6  CREATE TABLE trials (
7      nct_number VARCHAR(20) PRIMARY KEY,
8      study_title TEXT,
9      study_url TEXT,
10     study_status VARCHAR(100),
11     conditions TEXT,
12     interventions TEXT,
13     sponsor VARCHAR(255),
```



```

14     collaborators TEXT,
15     phases VARCHAR(100),
16     enrollment INT,
17     study_type VARCHAR(100)
18 );
19
20 CREATE TABLE conditions (
21     condition_id INT AUTO_INCREMENT PRIMARY KEY,
22     nct_number VARCHAR(20),
23     condition_name VARCHAR(255),
24     FOREIGN KEY (nct_number) REFERENCES trials(nct_number)
25 );
26
27 CREATE TABLE interventions (
28     intervention_id INT AUTO_INCREMENT PRIMARY KEY,
29     nct_number VARCHAR(20),
30     intervention_type VARCHAR(100),
31     intervention_name VARCHAR(255),
32     FOREIGN KEY (nct_number) REFERENCES trials(nct_number)
33 );
34
35 CREATE TABLE outcomes (
36     outcome_id INT AUTO_INCREMENT PRIMARY KEY,
37     nct_number VARCHAR(20),
38     outcome_title TEXT,
39     outcome_description TEXT,
40     time_frame VARCHAR(255),
41     FOREIGN KEY (nct_number) REFERENCES trials(nct_number)
42 );
43
44 CREATE TABLE adverse_events (
45     event_id INT AUTO_INCREMENT PRIMARY KEY,
46     nct_number VARCHAR(20),
47     event_name VARCHAR(255),
48     arm_group VARCHAR(255),
49     affected_count INT,
50     total_count INT,
51     FOREIGN KEY (nct_number) REFERENCES trials(nct_number)
52 );
53
54 ALTER TABLE trials
55 ADD COLUMN acronym VARCHAR(100),
56 ADD COLUMN sex VARCHAR(50),
57 ADD COLUMN age VARCHAR(100),
58 ADD COLUMN funder_type VARCHAR(100),
59 ADD COLUMN study_design TEXT,
60 ADD COLUMN start_date VARCHAR(50),
61 ADD COLUMN completion_date VARCHAR(50),
62 ADD COLUMN results_first_posted VARCHAR(50),
63 ADD COLUMN locations TEXT;

```

```

64
65 SELECT * FROM trials LIMIT 10;
66
67 SELECT study_status, COUNT(*) AS total
68 FROM trials
69 GROUP BY study_status;
70
71 SELECT phases, COUNT(*) AS total
72 FROM trials
73 GROUP BY phases;
74
75 SELECT sponsor, COUNT(*) AS total
76 FROM trials
77 GROUP BY sponsor
78 ORDER BY total DESC;
79
80 SELECT AVG(enrollment) AS avg_enrollment
81 FROM trials;
82
83 SELECT condition_name, COUNT(*) AS total
84 FROM conditions
85 GROUP BY condition_name
86 ORDER BY total DESC
87 LIMIT 10;
88
89 SELECT intervention_type, COUNT(*) AS total
90 FROM interventions
91 GROUP BY intervention_type
92 ORDER BY total DESC;
93
94 SELECT phases, AVG(enrollment) AS avg_enrollment
95 FROM trials
96 GROUP BY phases;
97
98 SELECT study_title, enrollment
99 FROM trials
100 ORDER BY enrollment DESC
101 LIMIT 10;

```

Appendix C Full Python Code

Listing 2 Python ETL, visualization, NLP, and machine learning workflow

```

1 # Loading Libraries
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6 from sqlalchemy import create_engine

```

```

7
8 # MySQL Connector
9 engine = create_engine(
10     "mysql+pymysql://root:YOUR_PASSWORD@localhost/
      ucsf_clinical_trials"
11 )
12
13 # Loading Dataset
14 df = pd.read_csv("ctg-studies.csv")
15
16 # Rename Columns for MySQL
17 df = df.rename(columns={
18     "NCT Number": "nct_number",
19     "Study Title": "study_title",
20     "Study URL": "study_url",
21     "Acronym": "acronym",
22     "Study Status": "study_status",
23     "Conditions": "conditions",
24     "Interventions": "interventions",
25     "Primary Outcome Measures": "primary_outcome_measures",
26     "Secondary Outcome Measures": "secondary_outcome_measures",
27     "Other Outcome Measures": "other_outcome_measures",
28     "Sponsor": "sponsor",
29     "Collaborators": "collaborators",
30     "Sex": "sex",
31     "Age": "age",
32     "Phases": "phases",
33     "Enrollment": "enrollment",
34     "Funder Type": "funder_type",
35     "Study Type": "study_type",
36     "Study Design": "study_design",
37     "Start Date": "start_date",
38     "Completion Date": "completion_date",
39     "Results First Posted": "results_first_posted",
40     "Locations": "locations"
41 })
42
43 # Clean Enrollment
44 df["enrollment"] = pd.to_numeric(df["enrollment"], errors="coerce")
      .fillna(0).astype(int)
45
46 # Load main trials table
47 trials_df = df[[
48     "nct_number", "study_title", "study_url", "acronym",
49     "study_status", "sponsor", "collaborators",
50     "sex", "age", "phases", "enrollment",
51     "funder_type", "study_type", "study_design",
52     "start_date", "completion_date", "results_first_posted",
53     "locations"
54 ]]

```

```

55 trials_df.to_sql("trials", con=engine, if_exists="append", index=
56     False)
57
58 # Conditions Table
59 condition_rows = []
60 for _, row in df.iterrows():
61     if pd.notna(row["conditions"]):
62         for condition in str(row["conditions"]).split("|"):
63             condition_rows.append({
64                 "nct_number": row["nct_number"],
65                 "condition_name": condition.strip()
66             })
67
68 conditions_df = pd.DataFrame(condition_rows)
69 conditions_df.to_sql("conditions", con=engine, if_exists="append"
70     , index=False)
71
72 # Interventions Table
73 intervention_rows = []
74 for _, row in df.iterrows():
75     if pd.notna(row["interventions"]):
76         for intervention in str(row["interventions"]).split("|"):
77             if ":" in intervention:
78                 intervention_type, intervention_name =
79                     intervention.split(":", 1)
80             else:
81                 intervention_type = "UNKNOWN"
82                 intervention_name = intervention
83
84             intervention_rows.append({
85                 "nct_number": row["nct_number"],
86                 "intervention_type": intervention_type.strip(),
87                 "intervention_name": intervention_name.strip()
88             })
89
90 interventions_df = pd.DataFrame(intervention_rows)
91 interventions_df.to_sql("interventions", con=engine, if_exists="
92     append", index=False)
93
94 print("Data loaded successfully!")
95 print("Trials:", len(trials_df))
96 print("Conditions:", len(conditions_df))
97 print("Interventions:", len(interventions_df))
98
99 # Summary Statistics
100 query = """
SELECT enrollment, phases, study_type, funder_type
FROM trials
WHERE enrollment > 0;

```

```

101 """
102 df = pd.read_sql(query, engine)
103 print(df.describe())
104 print(df.groupby("phases")["enrollment"].agg(["count", "mean", "
      median", "max"]))
105
106 # Log-scale Enrollment Chart
107 query = """
108 SELECT enrollment
109 FROM trials
110 WHERE enrollment > 0;
111 """
112 df = pd.read_sql(query, engine)
113 df["log_enrollment"] = np.log1p(df["enrollment"])
114 plt.figure(figsize=(10,6))
115 plt.hist(df["log_enrollment"], bins=20, color="lightgray",
      edgecolor="black")
116 plt.xlabel("Log Enrollment")
117 plt.ylabel("Frequency")
118 plt.title("Log-Transformed Distribution of Trial Enrollment")
119 plt.tight_layout()
120 plt.savefig("log_enrollment_distribution.png", dpi=300,
      bbox_inches="tight")
121 plt.show()
122
123 # Boxplot Enrollment by Phase
124 query = """
125 SELECT phases, enrollment
126 FROM trials
127 WHERE enrollment > 0;
128 """
129 df = pd.read_sql(query, engine)
130 plt.figure(figsize=(10,6))
131 df.boxplot(
132     column="enrollment",
133     by="phases",
134     patch_artist=True,
135     boxprops=dict(facecolor="lightgray", color="black"),
136     medianprops=dict(color="black")
137 )
138 plt.title("Enrollment Distribution by Trial Phase")
139 plt.suptitle("")
140 plt.xlabel("Trial Phase")
141 plt.ylabel("Enrollment")
142 plt.tight_layout()
143 plt.savefig("enrollment_boxplot_by_phase.png", dpi=300,
      bbox_inches="tight")
144 plt.show()
145
146 # Most Common Conditions

```

```

147 query = """
148 SELECT condition_name, COUNT(*) AS total
149 FROM conditions
150 GROUP BY condition_name
151 ORDER BY total DESC
152 LIMIT 10;
153 """
154 df = pd.read_sql(query, engine)
155 plt.figure(figsize=(14,6))
156 plt.bar(df["condition_name"], df["total"], color="lightgray",
157         edgecolor="black")
158 plt.xticks(rotation=60, ha="right")
159 plt.xlabel("Condition")
160 plt.ylabel("Number of Trials")
161 plt.title("Top Clinical Trial Conditions")
162 plt.tight_layout()
163 plt.savefig("top_conditions.png", dpi=300, bbox_inches="tight")
164 plt.show()
165
166 # Intervention Types
167 query = """
168 SELECT intervention_type, COUNT(*) AS total
169 FROM interventions
170 GROUP BY intervention_type
171 ORDER BY total DESC;
172 """
173 df = pd.read_sql(query, engine)
174 plt.figure(figsize=(8,5))
175 plt.bar(df["intervention_type"], df["total"], color="lightgray",
176         edgecolor="black")
177 plt.xticks(rotation=45, ha="right")
178 plt.xlabel("Intervention Type")
179 plt.ylabel("Count")
180 plt.title("Clinical Trial Intervention Types")
181 plt.tight_layout()
182 plt.savefig("intervention_types.png", dpi=300, bbox_inches="tight")
183 plt.show()
184
185 # Average Enrollment by Phase
186 query = """
187 SELECT phases, AVG(enrollment) AS avg_enrollment
188 FROM trials
189 GROUP BY phases;
190 """
191 df = pd.read_sql(query, engine)
192 plt.figure(figsize=(8,5))
193 plt.bar(df["phases"], df["avg_enrollment"], color="lightgray",
194         edgecolor="black")
195 plt.xlabel("Clinical Trial Phase")

```

```

193 plt.ylabel("Average Enrollment")
194 plt.title("Average Enrollment by Trial Phase")
195 plt.tight_layout()
196 plt.savefig("avg_enrollment_by_phase.png", dpi=300, bbox_inches="
    tight")
197 plt.show()
198
199 # Top Sponsors
200 query = """
201 SELECT sponsor, COUNT(*) AS total
202 FROM trials
203 GROUP BY sponsor
204 ORDER BY total DESC
205 LIMIT 10;
206 """
207 df = pd.read_sql(query, engine)
208 plt.figure(figsize=(12,6))
209 plt.barh(df["sponsor"], df["total"], color="lightgray", edgecolor
    ="black")
210 plt.xlabel("Number of Trials")
211 plt.ylabel("Sponsor")
212 plt.title("Top Clinical Trial Sponsors")
213 plt.tight_layout()
214 plt.savefig("top_sponsors.png", dpi=300, bbox_inches="tight")
215 plt.show()
216
217 # Enrollment Distribution
218 query = """
219 SELECT enrollment
220 FROM trials
221 WHERE enrollment > 0;
222 """
223 df = pd.read_sql(query, engine)
224 plt.figure(figsize=(10,6))
225 plt.hist(df["enrollment"], bins=20, color="lightgray", edgecolor=
    "black")
226 plt.xlabel("Enrollment")
227 plt.ylabel("Frequency")
228 plt.title("Distribution of Clinical Trial Enrollment")
229 plt.tight_layout()
230 plt.savefig("enrollment_distribution.png", dpi=300, bbox_inches="
    tight")
231 plt.show()
232
233 # Logistic Regression: Predict Trial Phase
234 from sklearn.model_selection import train_test_split
235 from sklearn.feature_extraction.text import TfidfVectorizer
236 from sklearn.linear_model import LogisticRegression
237 from sklearn.metrics import classification_report
238

```

```

239 query = """
240 SELECT study_title, phases
241 FROM trials
242 WHERE phases IS NOT NULL;
243 """
244
245 df = pd.read_sql(query, engine)
246 df = df.dropna()
247
248 X = df["study_title"]
249 y = df["phases"]
250
251 vectorizer = TfidfVectorizer(stop_words="english", max_features
    =500)
252 X_vectorized = vectorizer.fit_transform(X)
253
254 X_train, X_test, y_train, y_test = train_test_split(
255     X_vectorized,
256     y,
257     test_size=0.25,
258     random_state=42
259 )
260
261 model = LogisticRegression(max_iter=1000)
262 model.fit(X_train, y_train)
263
264 predictions = model.predict(X_test)
265 print(classification_report(y_test, predictions, zero_division=0)
    )
266
267 # K-Means Clustering Clinical Trials by Title
268 from sklearn.cluster import KMeans
269 from sklearn.feature_extraction.text import TfidfVectorizer
270
271 query = """
272 SELECT nct_number, study_title
273 FROM trials;
274 """
275
276 df = pd.read_sql(query, engine)
277 df = df.dropna()
278
279 vectorizer = TfidfVectorizer(stop_words="english", max_features
    =500)
280 X = vectorizer.fit_transform(df["study_title"])
281
282 kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
283 df["cluster"] = kmeans.fit_predict(X)
284
285 print(df[["nct_number", "study_title", "cluster"]].head(20))

```



```

286 df[["nct_number", "study_title", "cluster"]].to_csv("
287     kmeans_trial_clusters.csv", index=False)
288
289 # Convert sparse TF-IDF matrix to dense
290 X_dense = X.toarray()
291
292 # Reduce TF-IDF features to 2D
293 pca = PCA(n_components=2, random_state=42)
294 X_pca = pca.fit_transform(X_dense)
295
296 # Add PCA coordinates to dataframe
297 df["pca1"] = X_pca[:, 0]
298 df["pca2"] = X_pca[:, 1]
299
300 # Plot clusters
301 plt.figure(figsize=(10, 7))
302
303 plt.scatter(
304     df["pca1"],
305     df["pca2"],
306     c=df["cluster"],
307     cmap="viridis",
308     edgecolor="black",
309     alpha=0.8
310 )
311
312 plt.xlabel("Principal Component 1")
313 plt.ylabel("Principal Component 2")
314 plt.title("K-Means Clustering of Clinical Trial Titles")
315
316 plt.colorbar(label="Cluster")
317 plt.tight_layout()

```

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